QP Code: _____ **Reg. No: _____**
Name: _____

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) MODEL QUESTION PAPER
FIFTH SEMESTER BCA HONOURS EXAMINATION
Core Course
MG5CCRBCA300 – Computer Networks
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50 marks**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A
Short Answer Questions
Answer all questions. Each question carries 2 marks.

1. What is meant by resource sharing in computer networks? [K]/ [CO1]
2. Differentiate LAN and WAN. [K] / [CO1]
3. Briefly explain simplex, half duplex and full duplex transmission. [U] / [CO1]
4. Explain the functions of the router. [U] / [CO1]
5. Explain TCP/IP reference model. [K] / [CO2]
6. What is meant by byte stuffing? [U] / [CO2]
7. How many bits are used in an IPv4 address? Explain the address space in IPV4. [A] / [CO2]
8. Explain handshaking in TCP. [A] / [CO3]
9. What is a VPN and why it is used? [U] / [CO3]
10. How SDN is different from traditional networking? [A] / [CO3]

[10 × 2 = 20]

PART B
Short Essay Questions
Answer any 3 out of 5 questions. Each question carries 5 marks.

11. Explain various network topologies. [U] / [CO1]
12. Explain Circuit switching and Packet switching. Differentiate Datagram Networks and Virtual Circuit Networks. [U] / [CO1]
13. Explain functions of each layer in OSI reference model with a neat diagram [K] / [CO2]
14. Explain ALOHA, CSMA/CD and CSMA/CA protocols. [U] / [CO2]
15. Explain various congestion control techniques in TCP. [U] / [CO3]

[3× 5 = 15]

PART C
Essay Questions

Answer any 1out of 2 questions. Each question carries 15 marks

16. Explain the role of data link layer in error detection and error correction. [U] / [CO2]
17. Explain SNMP in detail. Describe its components, working and applications in network management. [A] / [CO3]

[1 × 15 = 15]